

ME1103 – Principles of Mechanics and Materials
MATERIAL PROPERTIES FROM TENSILE TESTS

VENUE: E3-04-02E

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NOTE: Wear proper attire in the laboratory. No sandals, slippers or any non-covered shoes are allowed.

1) INTRODUCTION

Investigation of the behavior of solids under external loads began with Galileo Galilei (164-1642), though Robert Hooke (1635-1703) was the first to point out that a body is deformed if a force acts upon it. Since then, numerous studies have been developed to the knowledge of the field of strength of materials.

Strength of materials (also called mechanics of materials or mechanics of deformable bodies) is the branch of applied mechanics that deals with internal behavior of loaded solid bodies. It focuses primarily on stress-strain analysis and on the mechanical properties of materials. Stress and strain are fundamental quantities connecting the internal forces to the deformations induced in the body, via its mechanical material properties. They are defined as the intensity of internal forces and deformation, respectively, in the body induced by the external loads imposed on it. By looking at the relationship between the stress and strain quantities, we can identify unique mechanical material properties that quantify the strength, ductility and toughness of the material.

2) OBJECTIVES

- Learn how to account for internal forces and corresponding deformation through the concepts of stress and strain.
- Experience carrying out a simplified tensile test and learning how to obtain many useful mechanical proper6es (e.g. stiffness, strength, ductility and toughness) from the test.

3) TENSILE TEST ON SILICONE RUBBER

A tensile test is one of the most useful and effective laboratory tests for obtaining several common mechanical proper6es. The mechanical properties of a material reflect the different ways in which the material responds to an applied force or load.

3.1) *Experimental Setup*

A [test stand with automatic servo control](#) (Fig. 1) is equipped with upper and lower grippers to hold the specimen. The upper gripper is connected to a force gauge that will record the force experienced during the tensile test. Vertical motion of the upper gripper is executed through the control panel on the test stand.

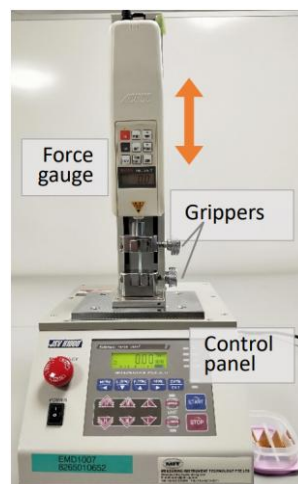


Fig.1 Benchtop force gauge tensile tester.

3.2) Test Specimens

Silicone rubber sheets are cut into dog-bone shapes with two wider ends for clamping between the grippers, and they have narrower gauge length in between, as shown in Fig. 2. This design ensures that the elongation of the test specimen is largely coming from the narrow portion. This implies that a more accurate estimation of the strain can be determined for this portion. Four specimens are to be tested, with two different gauge lengths and widths, but with the same thickness of 3 mm. The dimensions of the specimens are tabulated in Table 1.

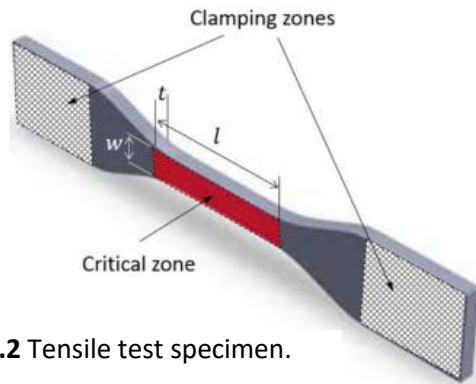


Fig.2 Tensile test specimen.

Table 1 Dimensions of test specimens within the gauge length
(thickness, $t = 3\text{ mm}$ in all cases)

Specimen No.	Gauge Length, l (mm)	Width, w (mm)
1	20	8
2	20	10
3	25	8
4	25	10

3.3) Experimental Procedure

1. Insert the specimen between the teeth of both grippers (Fig. 3a).
2. With the specimen seated at the base of the lower gripper, tighten the lower gripper.
3. Using the downward arrow buttons “M” or “L” on the control panel (Fig. 3b), lower the upper gripper until the force gauge just begins to register a force (Fig. 3c).
4. Using the upward “L” arrow button, move the upper gripper away until the force gauge reads zero again. Tighten the upper gripper. **Ensure the specimen is not twisted.**
5. **Zero the force gauge** if it reads a force (Fig. 3c). In addition, **press and hold the D/ZERO button on the control panel** to zero the displacement reading (Fig. 3b).
6. Put the plastic safety guard around the gripper assembly.
7. Pull the specimen in steps using the upward “M” arrow button. Each press of the “M” button will elongate the specimen by 0.5 mm. You can read the displacement on the screen of the control panel (Fig. 3b).
8. Take a force reading **starting from an elongation of 1 mm, in intervals of 1 mm** (press the “M” button twice), **until 5 mm**. Use the table in your log sheet to record your data, or record directly into an Excel work sheet.
9. Allow the reading on the force gauge to stabilize before recording the force.
10. Once the 5 mm elongation has been reached, return the upper gripper to its original position using the downward “H” or “M” arrow buttons.
11. Repeat the above steps for the other specimens.



Fig.3a Upper and lower grippers with specimen



Fig.3b Control panel



Fig.3c Force gauge

3.4) Results and Discussion

1. Plot the **force vs elongation data** for the 4 specimens onto the same graph.
2. **Convert** the **force and elongation** data into **stress and strain**. **Consolidate** all 4 sets of stress-strain data and plot a single **stress vs strain** graph, with a **line of best fit**.
3. When deformation is relatively small, most materials will return to their original shape and size when the load is removed. Such elastic deformation is non-permanent and reversible, exhibiting a linear relationship between stress and strain. This is known as Hooke's law:

$$\sigma = E\epsilon, \text{ where } E \text{ is the modulus of elasticity or Young's modulus (Pa)}$$

E may be thought of as stiffness (whether rigid or floppy), or a material's resistance to elastic deformation. A stiff material has a high E , and its deformation (strain) under an elastic load is small.

4. From the equation for the line of best fit in your stress-strain graph, **estimate the Young's modulus of elasticity** for silicone rubber.
5. Comment on the **effects of specimen dimensions** on the above results. Should specimen dimensions be standardized in tensile testing? (Hint: consider the force-elongation plots)

4) STRENGTH, DUCTILITY AND TOUGHNESS

Tensile tests are generally carried out until the specimen breaks. At some point after the initial elastic deformation, the stress-strain behaviour becomes non-linear and no longer obeys Hooke's law. The specimen does not return to its original length when the load is removed, although there is recovery of the small elastic component (sometimes called spring-back); this plastic deformation is now permanent and non-reversible. You are given an Excel file that contains stress and strain data from a tensile test of metallic alloy under [international standard testing conditions](#) on a commercial tensile tester.

4.1) Data plot and analysis

1. Plot a **graph of stress vs strain** from the data given.
2. Estimate the **Young's modulus of elasticity** for this metallic alloy from your graph. (Pause and think: which section of the curve most accurately reflects elastic deformation?)

- The point at which plastic deformation begins is known as **yielding**. Knowing the stress at which yielding occurs is vital for design, because most components are sized such that they should not yield (i.e. deform permanently) during operation.

In most materials, the transition from elastic to plastic deformation is gradual, making it difficult to determine the yield point precisely. The convention is to specify the yield strength, σ_y , as the stress at which a noticeable yet safe amount of plastic deformation has occurred, which, for metals, is usually a strain of 0.2% (0.002). This stress is known then as the **0.2% offset yield strength** or the 0.2% proof stress.

Estimate the **0.2% offset yield strength** for your alloy.

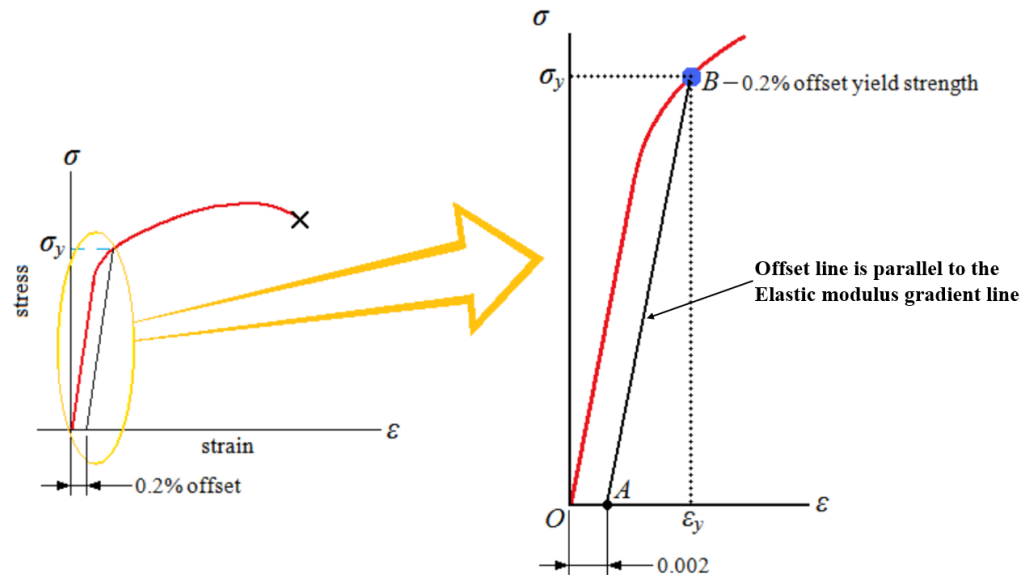


Fig.4 Estimating the proof stress from tensile test.

- The **ultimate tensile strength** (UTS) or sometimes, simply tensile strength (yes, it can be confusing!), is the maximum stress on the stress-strain curve. Most materials will fail quickly after the UTS has been reached. UTS is not as useful in design as the yield strength. However, for metal forming operations, e.g. bending, extrusion, drawing, it is widely used, as UTS is the limit before the onset of failure. **Estimate the UTS** from your graph.
- Ductility is a measure of the amount of plastic deformation that has been sustained at fracture. Ductility may be expressed as either percent elongation or percent reduction in area. Both involve fitting the two broken ends of the tensile specimen back together, measuring the final elongation or final diameter, and calculating the change in length or area as a percentage of the initial dimensions.

In the absence of actual broken tensile specimens, ductility in % elongation may be estimated from the stress-strain graph:

$$\% \text{ elongation} = \frac{\text{final gauge length} - \text{original gauge length}}{\text{original gauge length}} \times 100\%$$

Since strain, $\epsilon = \frac{\Delta l}{l_0}$, % elongation is simply the strain at fracture. Finally, after removing the elastic recovery (*spring-back*) which can be approximated from the green line (parallel to elastic modulus line) as shown in Fig.5, **estimate the % elongation** of the permanent deformation from your graph.

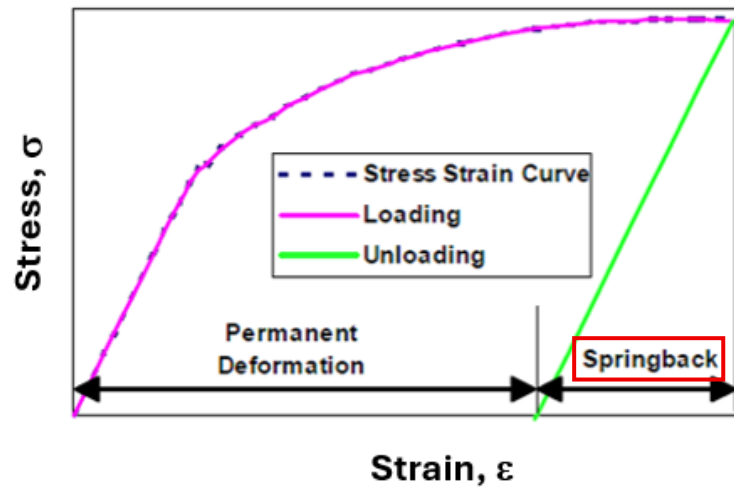


Fig.5 Estimating the elastic spring-back from tensile test.

6. One measure of **toughness** is the amount of energy absorbed by the material up to fracture, as depicted in Fig. 6. In the tensile test, this is given by the area under the stress-strain curve. **Estimate the toughness** using the [trapezoidal rule in Excel](#).

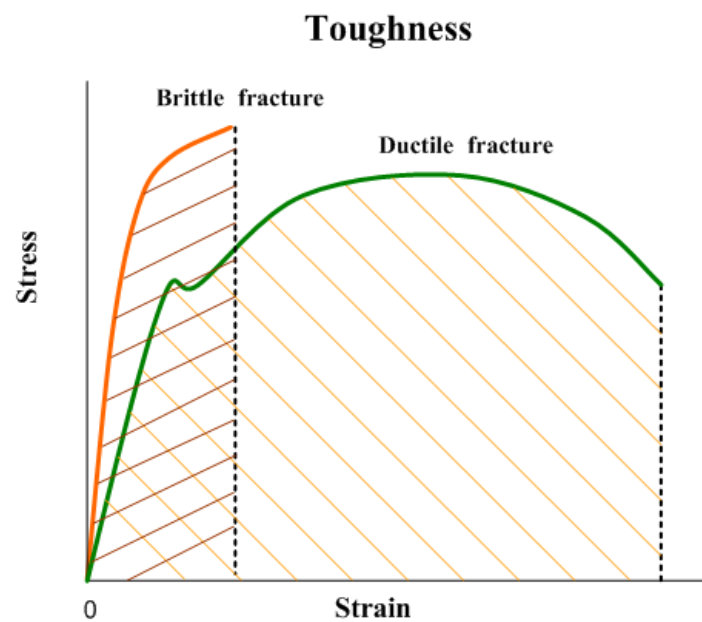


Fig.6 Fracture toughness of a material corresponds to the area under the stress-strain curve.

Estimate the toughness using the [trapezoidal rule in Excel](#).